AI-Powered Fleet Performance & Delivery Efficiency Dashboard

Project: Power BI Module End Project

Case Study Title: Fleet Performance & Delivery Efficiency Dashboard

Scenario:

A logistics company wants to analyze its fleet performance in terms of on-time deliveries, fuel efficiency, and cost per mile.

Dataset Fields:

- Trip data: Trip ID, Vehicle ID, Driver ID, Origin, Destination, Distance (km), Fuel Consumed (Liters), Delivery Status (On-Time/Late), Delivery Date.
- Vehicle Master: Vehicle ID, Vehicle Type, Capacity, Maintenance Cost.

Project Overview

The **Fleet Performance & Delivery Efficiency Dashboard** was developed using Power BI to analyze and monitor the operational efficiency of a logistics company's fleet. The primary objective of this project is to evaluate delivery performance, fuel consumption patterns, and transportation costs to support data-driven decision-making.

The dashboard integrates data from two key sources: Trip Data and Vehicle Master Data. Trip data provides insights into delivery status, distance travelled, fuel consumption, and trip-level details, while vehicle master data includes vehicle type, capacity, and maintenance cost information. By combining these datasets, the dashboard delivers a comprehensive view of fleet operations.

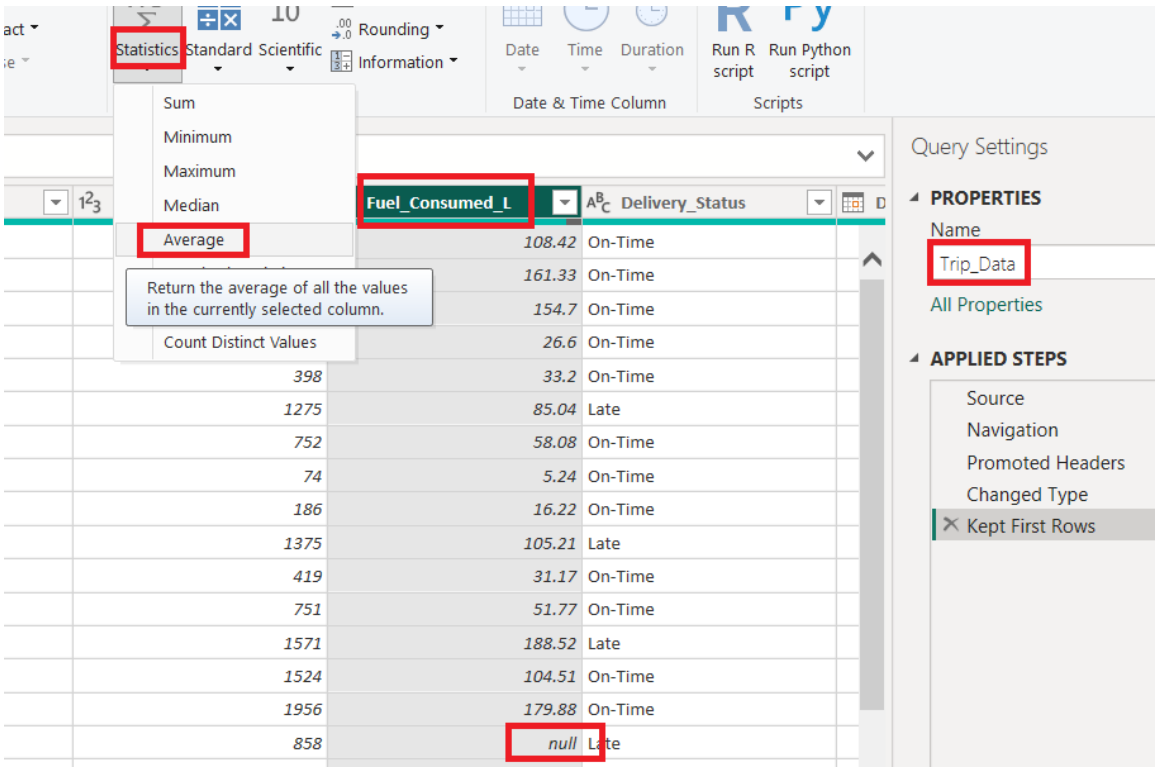
This project focuses on analyzing on-time delivery performance across destinations, monitoring fuel efficiency trends, evaluating vehicle utilization, and comparing maintenance and cost metrics by vehicle type. The insights generated help identify operational inefficiencies, optimize fleet allocation, reduce fuel and maintenance costs, and improve overall delivery reliability.

Overall, the dashboard serves as a strategic tool for improving fleet performance, enhancing cost control, and supporting efficient logistics management.

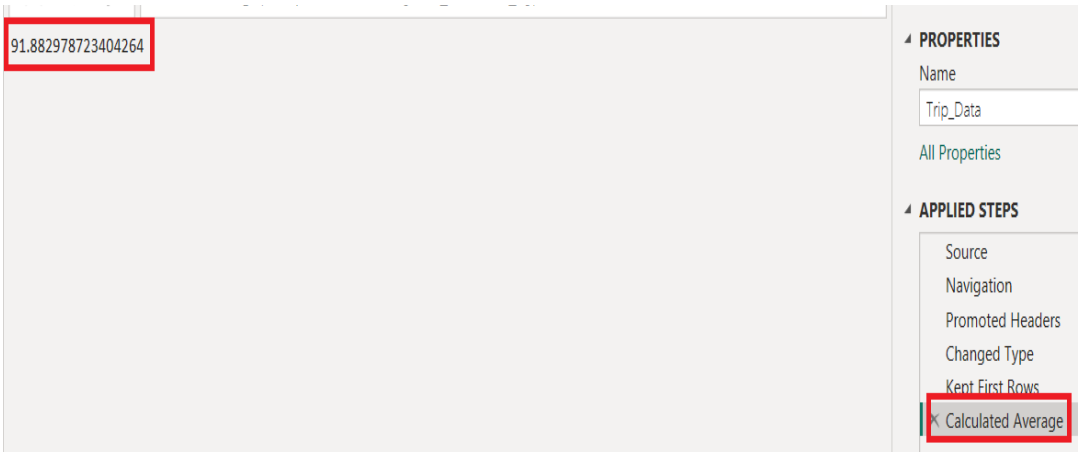
Tasks

Data Cleaning:

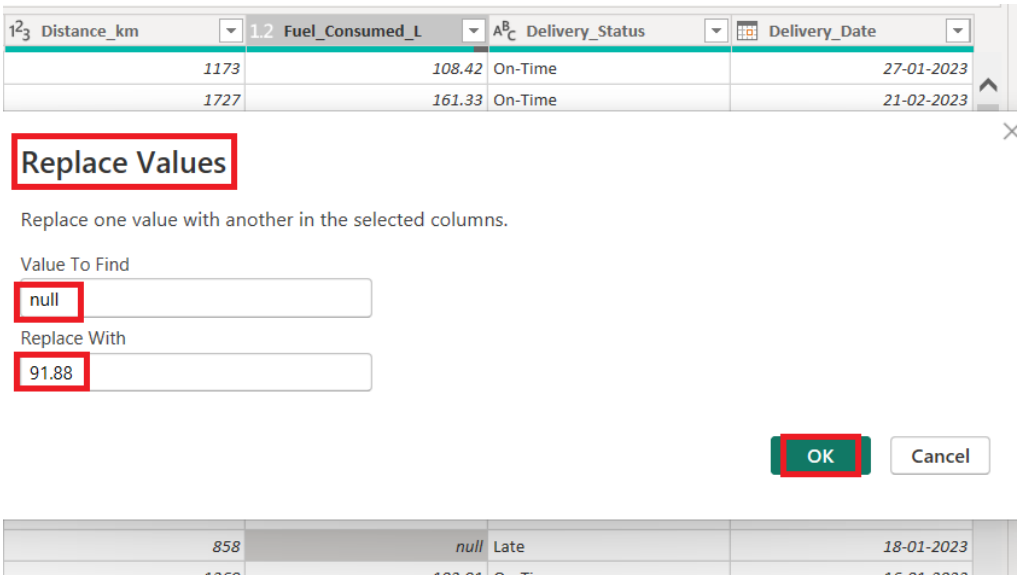
- In “Trip_Data” Table, there is a null value found in “Fuel_Consumed_L” column and it was filled by “Mean” imputation method by using Power BI-Power Query Editor.



- The Mean is calculated and the value is “91.88”

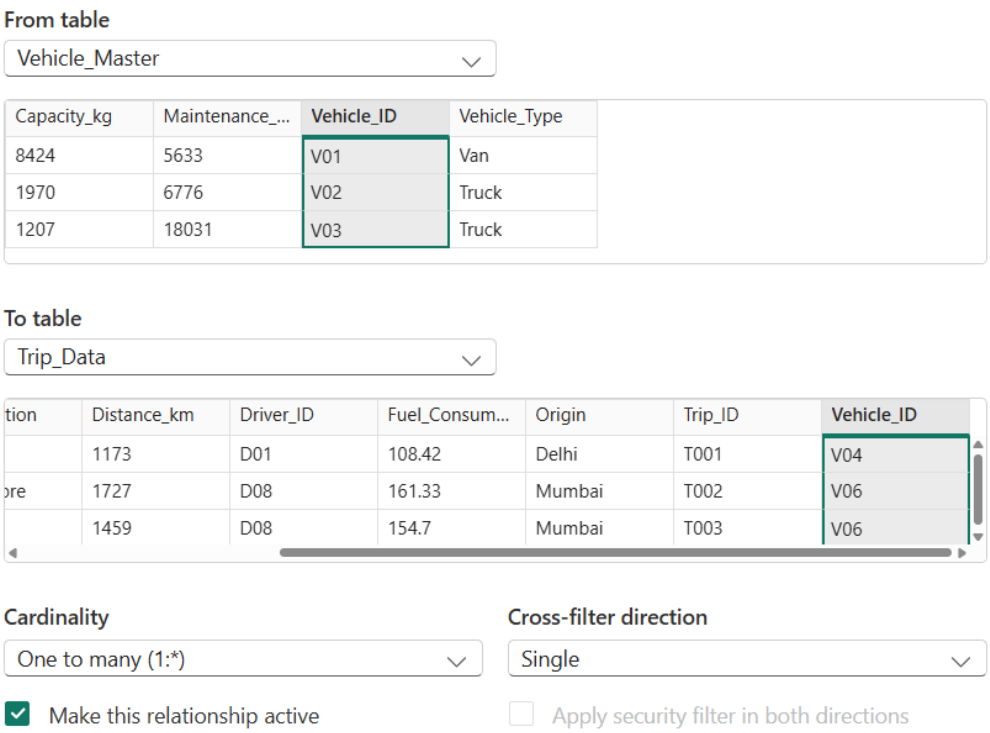


- The mean value is imputed on null values, by clicking “Replace values” and now the missing values are filled.

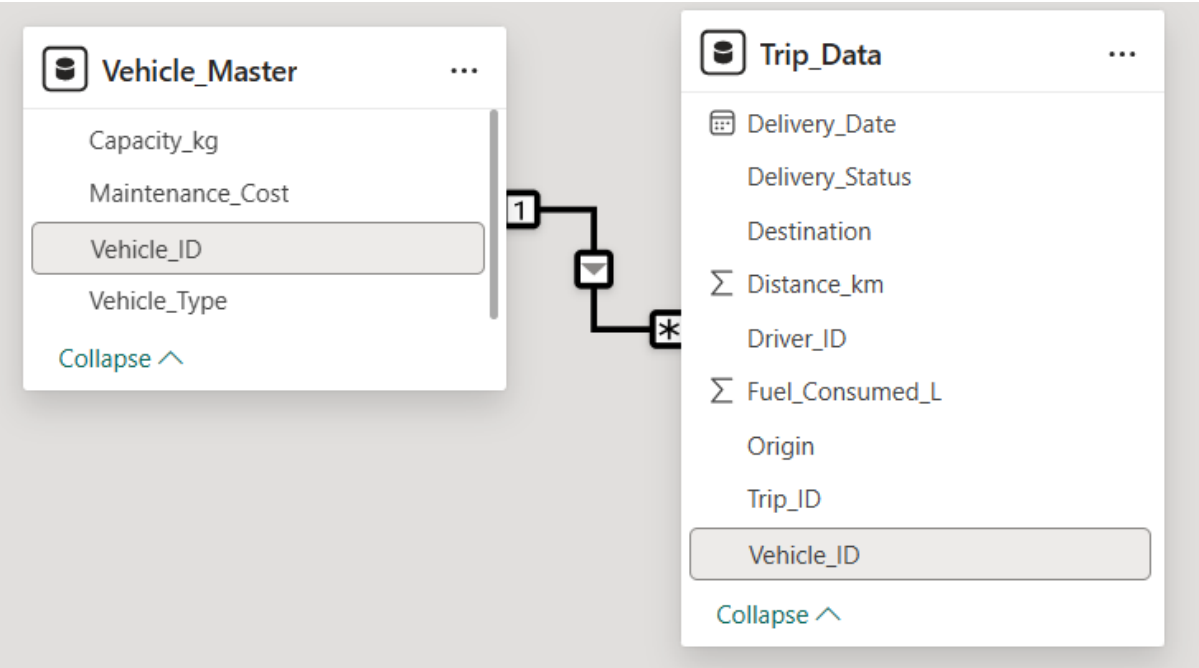


Data Modelling:

- To create a relationship using Vehicle_ID between Trip_Data and the Vehicle_Master table, go to “Model view”, click the “Home” tab, and click “Manage Relationship” where click “New relationship” and created the relationship as shown below.



- The relationship was created as shown in the image,

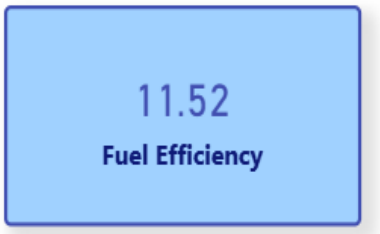


DAX Measures:

Fuel Efficiency:

- Created a DAX Measure for, **Fuel Efficiency = Distance / Fuel Consumed**, by using the following formula;

```
Fuel Efficiency = Divide(sum(Trip_Data[Distance_km]),sum(Trip_Data[Fuel_Consumed_L]))
```



On-Time Delivery %:

- Created DAX Measure for, **On-Time Delivery % = On-Time Trips / Total Trips**, to do this, calculated the number of "On-Time_Trips" and the "Total _Trips" in the Trip_Data table, and then divide both results.

```
On-Time_Trips = CALCULATE(COUNTROWS(Trip_Data),Trip_Data[Delivery_Status]="On-Time")
```

30
On-Time_Trips

```
Total_Trips = COUNTROWS(Trip_Data)
```

50
Total_Trips

- Here is the result for On-Time Delivery, by using the above two measures;

```
On-Time Delivery% = DIVIDE([On-Time_Trips],[Total_Trips])
```

30
On-Time_Trips

50
Total_Trips

0.60
On-Time Delivery%

Cost per km:

- Created DAX Measure for, **Cost per km = (fuel cost + Maintenance Cost) / Distance, where** fuel cost =100, for calculating this first find the Total_fuel_cost

```
Total_Fuel_Cost = sum(Trip_Data[Fuel_Consumed_L])*100
```

459.41K
Total_Fuel_Cost

- And then calculated the “cost per km” by using the following formula,

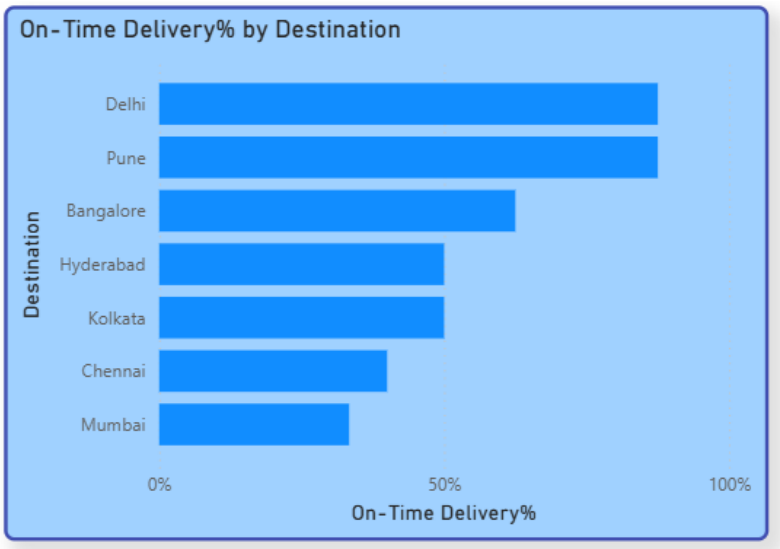
```
Cost per km = DIVIDE([Total_Fuel_Cost]+ sum(Vehicle_Master[Maintenance_Cost]),sum(Trip_Data[Distance_km]))
```

Visualization:

Bar chart: On-Time Delivery % by Destination

Insight:

- **Delhi** and **Pune** show the highest on-time delivery performance (close to 90–95%).
- **Mumbai** and **Chennai** show comparatively lower on-time percentages.
- There is a clear performance gap between top-performing and low-performing cities.
- Need Route optimization on low performing cities.

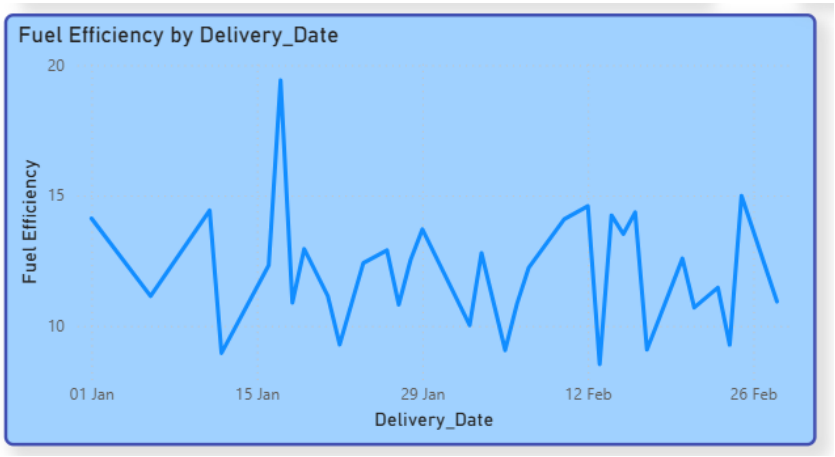


Line chart: Fuel efficiency trend by delivery date

Insight:

- Fuel efficiency fluctuates significantly across dates.

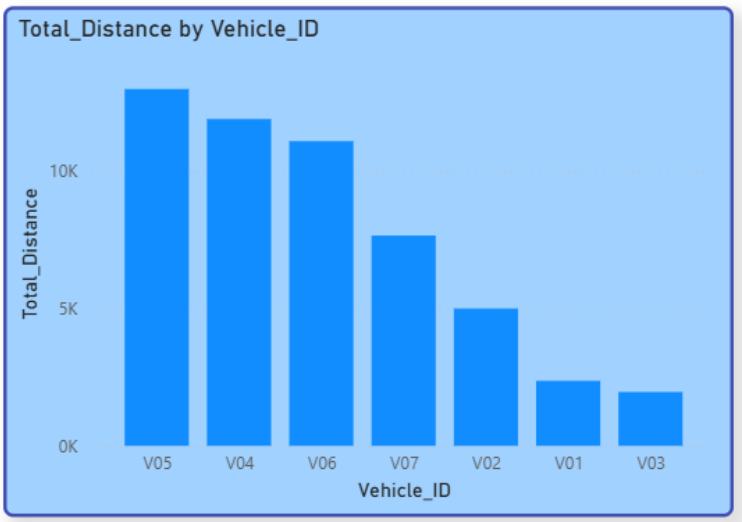
- There are sudden drops on certain days.
- Some days show unusually high efficiency peaks.



Column chart: Total_Distance by Each Vehicle

Insight:

- Vehicles **V05, V04, and V06** are used more heavily.
- Vehicles **V01 and V03** are underutilized.
- There is uneven workload distribution among vehicles.



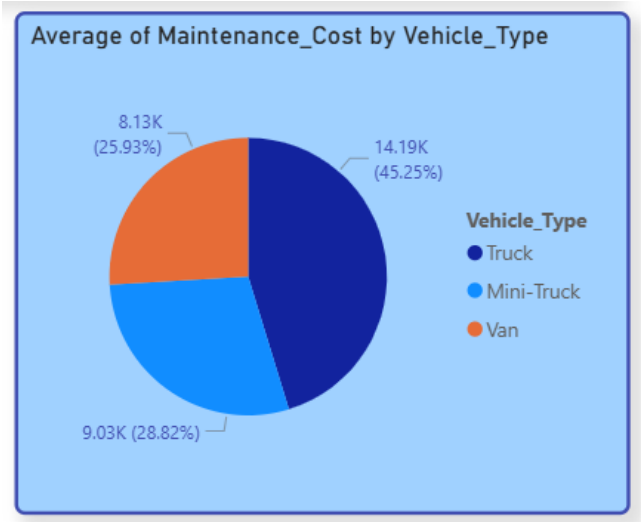
Cards visualization: Average cost per km

- A card visual was created for Average_Cost per km

Pie chart: vehicle type vs Average maintenance cost

Insight:

- **Trucks** have the highest maintenance cost share.
- **Mini-Trucks and Vans** have lower maintenance costs.
- Trucks contribute nearly half of the maintenance expense.



AI-Powered Visuals:

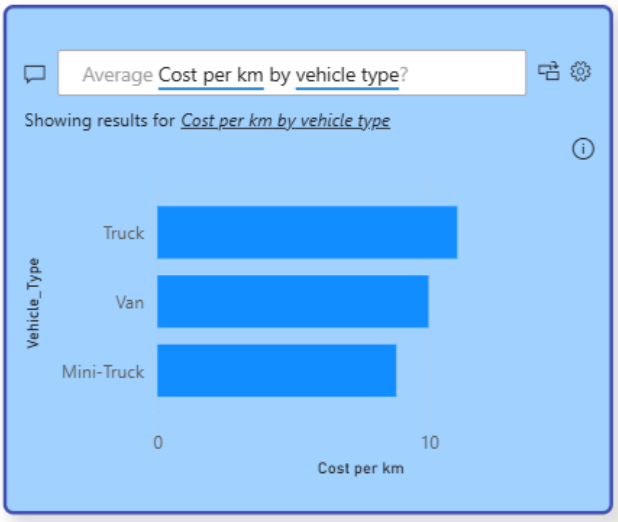
Q&A Visual:

Added Q&A visual → Prompt: "Average Cost per km by vehicle type?"

Insight:

- **Trucks** have the **highest** cost per km.
- Mini-Trucks have the lowest cost per km.

- Vans fall in between.

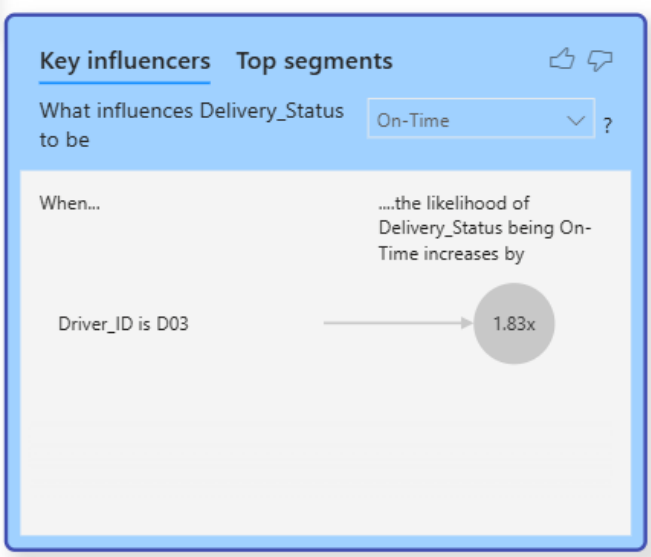


Key Influencers Visual:

Add Delivery Status in the Analyze and explain by -distance in km, vehicle_type, Driver_ID

Insight:

- The data identifies **Driver D03** as a "Key Influencer."
- When this specific driver is assigned, the likelihood of the delivery being on time increases by **1.83x** compared to all other drivers.



Decomposition Tree (AI Visual):

Analyze Cost per km → explained by Vehicle Type, Maintenance_cost, and Distance_km.

Insight:

- This chart tracks what drives the **10.11 Average Cost per km**.
- It shows that when **Maintenance Costs** hit the high bracket of **18,031**, the cost per km spikes to **18.84** (nearly double the average).
- It further breaks this down by **Distance**, showing that shorter distances often correlate with these higher costs per km.

Dashboard:

Key Insights:

- Fuel Efficiency trended down, resulting in a **22.60% decrease** between **Sunday, January 1, 2023 and Tuesday, February 28, 2023**.
- Fuel Efficiency started trending up on Tuesday, **February 21, 2023, rising by 2.11% in 7 days.**
- Fuel Efficiency **dropped** from **14.24 to 12.59** during its steepest decline between Tuesday, February 14, 2023 and Monday, February 20, 2023.
- **Pune** and **Delhi** tied for **highest On-Time Delivery% at 0.88**, followed by Bangalore. Mumbai had the lowest On-Time Delivery% at 0.33.
- Across all 7 Destination, **On-Time Delivery%** ranged from **0.33 to 0.88**.
- **V05** accounted for **24.53% of Total_Distance**.
- **Trucks** account for the largest share of maintenance expenses at **45.25% (14.19K)**, whereas **Vans** have the lowest maintenance impact at **25.93%**.
- **Driver ID D03** is the top influencer for reliability, increasing the likelihood of an **On-Time** delivery status by **1.83x**.
- **Vehicle V05** shows the highest utilization with **13.0K km** travelled, while **Vehicle V03** has the lowest operational footprint at only **2.0K km**.

Overall Conclusion

- Delivery performance varies across destinations, indicating the need for route-level improvements.
- Fuel efficiency is inconsistent across dates, highlighting opportunities for better fuel management.
- Fleet utilization is uneven, with some vehicles overused and others underutilized.
- Trucks contribute higher maintenance and operational costs compared to smaller vehicles.
- Vehicle type selection significantly impacts overall transportation cost and profitability.

Overall, the dashboard provides actionable insights to improve delivery reliability, optimize fleet usage, control operational costs, and enhance overall transport efficiency.